Hypoglycemia

Assess Blood Glucose

Hypoglycemia
RBS <70 mg/dl

Oral Glucose

EMR

	ADULT	PEDIATRIC
Oral Glucose	15g (1 tube)	½ tube *adjust to patient weight

OR

Glucose 10% IV/IO

EMT

	ADULT	PEDIATRIC
10% Glucose IV/IO	125ml – 250ml	10ml/kg

OR

Glucagon IM

EMT

	ADULT	PEDIATRIC
Glucagon IM	1mg	<6 years = 0.5mg
		>6 years = 1mg

Reassess

- Repeat RBS if altered mental status persists and treat accordingly
- **Look for other causes of altered mental status if maximal doses of dextrose do not improve mental status**

Consider Tele EMS

Consider Red Flags

- Severe hypoglycemia (RBS <70 mg/dL with seizures, unresponsiveness, or coma).
- Hypotension (SBP <90 mmHg in adults, or age-specific for pediatrics) or signs of shock.
- Persistent symptoms despite initial interventions.
- Persistent altered mental status after treatment.
- Bradycardia, or shock

Patient Care Goals

1. Rapidly identify and manage hypoglycemia to prevent neurological injury.
2. Restore blood glucose levels to a safe range.
3. Prevent hypoglycemia recurrence and ensure appropriate disposition.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients presenting with symptoms suggestive of hypoglycemia, including:
  - Confusion, irritability, sweating, shakiness, seizures, or unresponsiveness; random blood sugar (RBS) <70 mg/dL.

Exclusion Criteria

- Hyperglycemia without hypoglycemic symptoms.
- Altered mental status or symptoms clearly attributable to another cause (e.g., stroke, trauma, or drug overdose without glucose abnormality).
- Patients with normal blood glucose levels and no clinical suspicion of glycemic disturbance.

Key Patient Assessment Points

1. **Glucose Measurement**
   - Check RBS to confirm hypoglycemia (<70 mg/dL) .
   - Repeat RBS after interventions to monitor response.
2. **History Specific to Glycemic Disorders**
   - Confirm history of diabetes (type 1 or type 2), insulin use, or oral hypoglycemic agents.
   - Inquire about recent medication compliance (e.g., excess hypoglycemic agents).
   - Identify precipitating factors (e.g., infection, stress, dietary changes).
3. **Symptoms of Hypoglycemia**
   - Assess for autonomic symptoms (e.g., sweating, tremors, anxiety, palpitations).
   - Evaluate neurological symptoms (e.g., confusion, irritability, seizures, or coma).
4. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.

Patient Management & Interventions

1. **Primary and Secondary Survey**
   - Perform primary and secondary survey as per Universal Care CPG.
   - Patients who are unresponsive appropriate airway assessment and appropriate management.
   - Unconscious patients should be placed in the recovery position as a minimum while treatment is being administered.
   - Consider Oxygen Administration:
     - Consider oxygen if symptoms of hypoxia or respiratory distress are present.
     - Titrate to maintain SpO2 ≥94%.
2. **Consider Red Flags**
   - Severe hypoglycemia (RBS <70 mg/dL with seizures, unresponsiveness, or coma).
   - Hypotension (SBP <90 mmHg in adults, or age-specific for pediatrics) or signs of shock.
   - Persistent symptoms despite initial interventions.
   - Persistent altered mental status after treatment.
   - Bradycardia, or shock.
3. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
4. **Establish IV Access**
   - For administration of IV dextrose or fluids.
   - Consider IO access if IV unsuccessful and patient is critical.
   - Avoid fluids in uncomplicated hypoglycemia unless hypotension is present.

- Monitor for fluid overload, especially in patients with cardiac or renal comorbidities.

5. **Fluid Therapy**
   - Avoid fluids in uncomplicated hypoglycemia unless hypotension is present.
   - Monitor for fluid overload, especially in patients with cardiac or renal comorbidities.
6. **Dextrose 10%**
   - Adults: 100-200 mL IV (10-20 g), repeat as needed if RBS remains <70 mg/dL.
   - Pediatrics: 2-4 mL/kg IV (0.2-0.4 g/kg, max 25 g).
7. **Glucose 40% Oral Paste**
   - Adults: 15-20 g orally if conscious and able to swallow.
   - Pediatrics: 0.5-1 g/kg orally (max 15 g).
8. **Glucagon**
   - Indications:
     - If IV access is unavailable and patient is unresponsive
   - Adults: 1 mg IM. Once only.
   - Pediatrics: 0.5 mg IM (for <20 kg) or 1 mg (for ≥20 kg). Once only.
9. **Patient Reassessment**
   - Frequently Reassess the patient as per Universal Care CPG.
   - Specific Patient Reassessment Points:
     - **Glucose Reassessment**
       - Recheck RBS after each intervention.

Patient Safety Considerations

- Ensure airway is secure before administering oral glucose.
- Monitor closely for seizures or vomiting.
- Reassess blood glucose after treatment to ensure response.
- Document all treatment and timing of reassessment.

Clinical Insights & Guidance

- Rule out alternative causes of altered mental status such as stroke, head trauma, or overdose.
- Consider hypoglycemia as a secondary complication in seizure patients.
- If family or caregivers administered treatment prior to EMS arrival, confirm substance and dose.
- Patients with repeated hypoglycemia should be advised to follow up with their diabetes care provider.

Documentation

- Identification: Full patient name, age, and gender.
- Mechanism / Medical Complaint: Suspected or known hypoglycemia; symptoms such as confusion, tremor, seizure.
- Injuries / Information: Diabetes history, insulin or oral agents, missed meals, recent illness.
- Signs: Pre- and post-treatment RBS, AVPU or GCS, vitals, seizure activity.
- Treatment:
  - Oral glucose, IV dextrose, or IM glucagon (dose, route, response).
  - Reassessment after treatment.
- Allergies: Known drug allergies relevant to treatment.
- Medications: Diabetic medications taken or missed.
- Background: Diabetes, renal disease, past hypoglycemia episodes.
- Other: Scene details, bystander interventions, hospital notification.

Key Performance Indicators

1. Perform and document RBS before and after treatment.
2. Administer glucose (as appropriate) for all patients with an RBS<70mg/dL.

References

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